ST02

Chemistry in Everyday Life

Chemistry supplies 5-7 of the 18-23 Science and Technology questions. Alloy composition and common-name-to-formula matching are documented repeats - the alloy list alone has been set as a matching question more than once. Acids with their natural sources, the pH cluster, and the CNG-LPG pair are the other reliable suppliers.

This chapter is a vocabulary test wearing a chemistry coat. Baking soda, washing soda, quicklime, blue vitriol, brass, bronze, german silver - RPSC picks a household name and asks for the formula, the composition or the use. There is very little to understand and a great deal to fix in memory, so these notes are built around three master tables: alloys, ores, and common chemical names.

Chemistry in Everyday Life

Matter
- Solid, liquid, gas + plasma + BEC
- Sublimation - camphor, naphthalene, iodine
- Mercury only liquid metal
- Bromine only liquid non-metal

Atom + table
- Thomson electron, Chadwick neutron
- Mendeleev = atomic mass
- Moseley = atomic number
- Fluorine most electronegative

Acids, bases, pH
- Formic - ant; acetic - vinegar
- Lactic - curd; oxalic - tomato
- pH 7 neutral, blood 7.35-7.45
- Tooth decay below pH 5.5

Corrosion
- Rust = Fe2O3.xH2O
- Needs oxygen AND moisture
- Galvanisation = zinc
- Kalai / tinning = tin

Alloys (repeat)
- Brass = Cu + Zn
- Bronze = Cu + Sn
- German silver = Cu + Zn + Ni
- Amalgam = anything + mercury

Carbon
- Diamond hardest, four bonds
- Graphite conducts, lubricates
- Fullerene C60 football
- Graphene, carbon nanotubes

Fuels
- CNG = methane
- LPG = butane + propane
- Hydrogen highest calorific value
- Water gas = CO + H2

Materials
- Thermoplastic vs thermosetting
- Soap - micelle, fails in hard water
- Cement + gypsum slows setting
- Vulcanisation = rubber + sulphur

Drugs + food
- Antiseptic living, disinfectant non-living
- Penicillin - Fleming 1928
- Argemone in mustard oil
- Metanil yellow in turmeric

States of matter - and the two extra states the exam adds

Matter exists in three familiar states because of a tug of war between the force holding particles together and the energy making them move. In a solid the force wins, so shape and volume are both fixed. In a liquid the two are balanced, so volume is fixed but shape is not. In a gas the energy wins and neither is fixed. Beyond these three, the exam expects two more names.

Changes of state – learn the name of each arrow

Solid – MELTING (fusion) – Liquid →

Liquid – VAPORISATION (boiling or evaporation) – Gas →

Gas – CONDENSATION – Liquid →

Liquid – FREEZING (solidification) – Solid →

Solid straight to Gas – SUBLIMATION

- Sublimation, solid straight to vapour with no liquid stage: camphor, naphthalene balls, iodine, ammonium chloride and dry ice.
- PLASMA is the fourth state – a hot ionised gas. It is what glows inside a fluorescent tube and a neon sign, and what stars are made of.
- The BOSE-EINSTEIN CONDENSATE is the fifth state, formed at temperatures close to absolute zero. Satyendra Nath Bose's work with Einstein gives it the Indian half of its name.
- Evaporation happens at every temperature and only from the surface; boiling happens throughout the liquid at a fixed temperature. Evaporation always causes cooling.
- Mercury is the ONLY metal liquid at room temperature; bromine is the ONLY non-metal liquid at room temperature.
- Sodium and potassium are so reactive that they are stored under kerosene; white phosphorus is stored under water.

TRAP

Do not confuse dry ice with ordinary ice. Dry ice is solid CARBON DIOXIDE – it sublimes, leaves no puddle, and keeps ice cream and vaccines cold. Heavy water, D2O, is a different thing again: water made with deuterium, used as a moderator in nuclear reactors.

Atomic structure and the periodic table - who found what, and the trends

The atom was pieced together over about forty years, and RPSC asks those discoveries as a matching question. Learn the three particles with their discoverers, then learn the two ways of arranging the elements - Mendeleev's old way and the modern way. That one distinction is the most asked point in this section.

Sub-atomic particles and their discoverers

Particle	Charge	Discovered by
Electron	Negative	J. J. Thomson (1897)
Proton	Positive	Goldstein found the canal rays; Rutherford identified and named the proton
Neutron	No charge	James Chadwick (1932)
The nucleus itself	Positive, tiny, extremely dense	Ernest Rutherford, gold foil experiment

- Atomic number = number of protons. Mass number = protons + neutrons.
- ISOTOPES - same atomic number, different mass number (hydrogen, deuterium, tritium; carbon-12 and carbon-14).
- ISOBARS - same mass number, different atomic number. Do not swap the two words; that swap is the standard trap.
- Bonding in one line: ionic bond = transfer of electrons (common salt); covalent bond = sharing of electrons (water, methane); metallic bond = a sea of free electrons, which is why metals conduct.
- Hydrogen bonding between water molecules gives water its unusually high boiling point and makes ice less dense than liquid water.

Mendeleev arranged the elements in order of increasing atomic MASS and left gaps for elements not yet discovered - eka-aluminium and eka-silicon, later found and named gallium and germanium. But a few pairs stubbornly sat in the wrong order. Moseley fixed that: the modern periodic law says properties are a periodic function of ATOMIC NUMBER. The modern table has 18 groups and 7 periods.

- Group 1 = alkali metals. Group 2 = alkaline earth metals. Group 17 = halogens. Group 18 = noble or inert gases.
- Across a period from left to right: atomic size decreases, ionisation energy and electronegativity increase, metallic character decreases.
- Down a group: atomic size increases, ionisation energy and electronegativity decrease, metallic character increases.
- FLUORINE is the most electronegative element. Caesium and francium are among the most electropositive.
- Metalloids sit on the staircase between metals and non-metals - boron, silicon, germanium, arsenic, antimony, tellurium.
- Graphite is the exception every paper likes: a non-metal that conducts electricity.

Periodic trends - which way does each one go?

Modern table - 18 groups, 7 periods

Atomic size
- Falls across a period
- Rises down a group

Ionisation energy
- Rises across a period
- Falls down a group

Electronegativity
- Rises across a period
- Falls down a group
- F is the highest

Metallic character
- Falls across a period
- Rises down a group

Acids, bases, salts and pH – the chemistry of your kitchen and your stomach

An acid gives hydrogen ions in water and tastes sour; a base gives hydroxide ions, tastes bitter and feels soapy. That is all the theory you need. What RPSC actually asks is which acid sits in which everyday thing – the ant that stings you, the curd that turned sour, the tamarind in the chutney. Learn the source list first, then the pH ladder.

Acid to natural source – a documented repeat as a matching question

Acid	Found in
Formic (methanoic) acid	Sting of an ant, a bee and the nettle plant
Acetic acid	Vinegar
Citric acid	Lemon, orange and other citrus fruit
Lactic acid	Curd and sour milk; also builds up in tired muscles
Oxalic acid	Tomato and spinach
Tartaric acid	Tamarind and grapes
Malic acid	Apple
Hydrochloric acid	Gastric juice in the stomach
Butyric acid	Rancid butter
Uric acid	Urine

- Indicators: litmus turns red in acid and blue in base. Methyl orange is red in acid, yellow in base. Phenolphthalein is colourless in acid, pink in base.
- Turmeric paste turns red with a base - which is why a haldi stain on a white shirt turns red the moment soap touches it.
- The pH scale runs 0 to 14. At 25 C, pH 7 is neutral, below 7 acidic, above 7 basic. One pH unit means a tenfold change in acidity.
- Human blood is held tightly at pH 7.35 to 7.45. Tooth decay begins when the pH in the mouth falls below 5.5, which is why toothpaste is mildly basic.
- Rain is called acid rain when its pH drops below about 5.6, caused by oxides of sulphur and nitrogen.
- A bee sting is acidic and is soothed with baking soda; a wasp sting is alkaline and is soothed with vinegar. Keep that pairing straight - it gets reversed in options.

The pH ladder of familiar things - use it for ordering questions

Substance	Approximate pH	Nature
Gastric juice	1 to 3	Strongly acidic
Lemon juice	About 2	Acidic
Vinegar	About 3	Acidic
Milk	About 6.5	Very slightly acidic
Pure water	7	Neutral
Human blood	7.35 - 7.45	Very slightly basic
Baking soda solution	About 9	Basic
Milk of magnesia	About 10	Basic
Sodium hydroxide solution	13 - 14	Strongly basic

The pH scale and the two thresholds that are asked

Strongly basic — pH 13–14
sodium hydroxide solution

Basic — pH 9–10
milk of magnesia 10, baking soda 9

Human blood — pH 7.35–7.45
held very slightly basic

pH 7 – NEUTRAL (pure water)

Weakly acidic — pH 6–7
milk about 6.5

pH 5.5 – tooth decay begins

Strongly acidic — pH 1–3
gastric juice, lemon 2, vinegar 3

- Salts come from an acid plus a base. Sodium chloride is neutral, ammonium chloride is acidic, sodium carbonate is basic.
- Baking soda, $NaHCO_3$ – an antacid, the raising agent in baking, and the powder in a soda-acid fire extinguisher. Baking powder is baking soda plus a mild edible acid such as tartaric acid.
- Washing soda, $Na_2CO_3.10H_2O$ – used in laundry, in glass and paper making, and to soften hard water.
- Chlor-alkali process: electrolysis of brine gives caustic soda in solution, chlorine at the anode and hydrogen at the cathode.
- Bleaching powder, $CaOCl_2$, is made by passing chlorine over dry slaked lime; it bleaches cloth and disinfects drinking water.
- Plaster of Paris, $CaSO_4.(1/2)H_2O$, is gypsum heated to about 100 C; it sets hard with water and is used for plaster casts and mouldings.
- Antacid logic: acidity is excess HCl in the stomach, so the cure must be a MILD base – milk of magnesia or baking soda, never a strong base.

ASKED IN RAS

Repeat cluster: 'tooth decay begins when the pH of the mouth falls below 5.5' is set again and again, usually paired with a false statement. The acid-to-source matching - formic with ant, lactic with curd, oxalic with tomato - is also a documented repeat.

Oxidation, reduction and corrosion – why iron rusts and silver goes black

Oxidation used to mean adding oxygen and reduction meant removing it. The wider definition is about electrons: oxidation is loss of electrons, reduction is gain. The two always happen together, which is why the reaction is called redox. Corrosion is simply metal being slowly oxidised by its surroundings, and rusting is the version that costs India the most money.

- Mnemonic: OIL RIG - Oxidation Is Loss of electrons, Reduction Is Gain of electrons.
- Rust is hydrated ferric oxide, $Fe_2O_3.xH_2O$. It is brownish and flaky and does not protect the metal underneath it.
- Rusting needs BOTH oxygen and moisture. Iron does not rust in dry air, nor in boiled water free of dissolved air. Salt and acid speed it up, which is why coastal iron rusts fastest.
- Rusting is a chemical change - the rust weighs more than the iron it came from.
- Other corrosion you should be able to name: silver tarnishes BLACK (silver sulphide) in polluted air; copper and bronze turn GREEN (basic copper carbonate); aluminium forms a thin oxide film that actually protects it.
- Reactivity series, most reactive first: K, Na, Ca, Mg, Al, Zn, Fe, Pb, (H), Cu, Hg, Ag, Au, Pt. A metal higher up displaces one lower down from its salt solution.
- Gold and platinum sit at the bottom, which is why they occur free in nature and do not corrode.

The reactivity series – most reactive at the top

K, Na, Ca
most reactive of all
Won only by electrolysis

Mg, Al, Zn
Zn is above Fe, so galvanising protects
Al from bauxite by electrolysis

Fe, Pb
rust and corrode readily
Iron: blast furnace with coke

HYDROGEN sits here

Cu, Hg, Ag
below hydrogen in the series
Cu from copper pyrites, Khetri

Au, Pt
least reactive; do not corrode
Found free; only aqua regia attacks

Preventing rust – method and the metal used

Method	What is done	Note
Painting, greasing, oiling	A physical barrier against air and water	Cheapest, but a single scratch breaks it
Galvanisation	Coating iron or steel with ZINC	Zinc is more reactive, so it corrodes first – sacrificial protection; it works even if the coating is scratched
Tinning (kalai)	Coating brass and copper utensils with a thin layer of TIN	Stops copper from contaminating the food; the traditional Indian kalai
Electroplating	Depositing chromium or nickel by electrolysis	Decorative as well as protective
Alloying	Adding chromium to make stainless steel	Chromium forms an invisible, adherent oxide film
Cathodic protection	Bolting blocks of zinc or magnesium to ships and pipelines	The block corrodes instead of the iron

ASKED IN RAS

Asked in RAS Pre 2018 - brass utensils are coated with a thin layer of TIN, the process called kalai, so that copper does not contaminate food. 'Rust is hydrated ferric oxide' is a documented repeat, and galvanisation-with-zinc usually appears alongside it.

TRAP

Galvanisation is ZINC; tinning or kalai is TIN. The two start the same way in your head and RPSC swaps them deliberately. And remember: stainless steel resists rust because of CHROMIUM, not because of nickel.

Metals, their ores and alloys - the biggest scoring table in this chapter

Metals are almost never found pure. They sit in the ground as oxides, sulphides and carbonates - the ores. Getting the metal out follows the same steps every time, and the pure metal is then usually mixed with another to improve it. That mixture is an alloy, and alloy composition is the most frequently repeated question in the whole chemistry syllabus. Alloying makes a metal harder, more corrosion-resistant, sometimes lower-melting (solder) and sometimes just better looking.

How a metal is extracted from its ore

Concentration - remove the earthy waste (gangue) →

Roasting for sulphide ores, calcination for carbonate ores, to get the oxide →

Reduction of the oxide to the metal - by carbon, or by electrolysis for very reactive metals such as aluminium →

Refining - electrolytic refining gives the pure metal

Ore to metal - straight matching, learn both columns

Ore	Metal	Note
Bauxite	Aluminium	Extracted by electrolysis, aluminium being highly reactive
Haematite (Fe_2O_3), Magnetite (Fe_3O_4)	Iron	Reduced in a blast furnace with coke and limestone
Galena (PbS)	Lead	A sulphide, so it is roasted
Cinnabar (HgS)	Mercury	-
Zinc blende (ZnS), Calamine ($ZnCO_3$)	Zinc	Rajasthan's Zawar and Rampura-Agucha are the national centres
Copper pyrites ($CuFeS_2$), Malachite	Copper	Khetri belt in Rajasthan
Pitchblende	Uranium	-
Monazite	Thorium	Kerala's beach sands
Argentite	Silver	-
Rock salt	Sodium	-

ALLOY MASTER TABLE part 1 - the single most repeated matching set

Alloy	Composition	Used for
Brass	Copper + Zinc	Utensils, fittings, musical instruments
Bronze	Copper + Tin	Statues, medals, coins

Alloy	Composition	Used for
German silver	Copper + Zinc + Nickel - contains NO silver at all	Utensils and ornamental ware
Stainless steel	Iron + Chromium (about 18%) + Nickel (about 8%) + Carbon	Cutlery, utensils, surgical instruments
Steel	Iron + Carbon in a small percentage	Construction, tools, rails
Duralumin	Aluminium + Copper + Magnesium + Manganese	Aircraft bodies - light and strong
Solder	Lead + Tin, roughly equal parts	Joining electrical wires; it has a low melting point

ALLOY MASTER TABLE part 2

Alloy	Composition	Used for
Gunmetal	Copper + Tin + Zinc	Gears, bearings, marine fittings
Bell metal	Copper + Tin, about 78 : 22	Temple bells and gongs
Nichrome	Nickel + Chromium + Iron	Heating elements in irons, toasters, heaters
Magnalium	Aluminium + Magnesium	Light structures
Invar	Iron + Nickel	Very low thermal expansion - clock pendulums, measuring tapes
Amalgam	ANY alloy that contains mercury	Dental filling; extraction of gold and silver
Gold jewellery	24 carat is pure gold; 22 carat is about 91.6% gold with copper or silver	Pure gold is too soft to wear, so it is alloyed

The same alloys grouped by their base metal

Alloys by base metal

Copper base — five of them
- Brass: + Zn
- Bronze: + Sn
- German silver: +Zn+Ni
- Gunmetal: +Sn+Zn
- Bell metal: 78Cu:22Sn

Iron base — the steels
- Steel: + C
- Stainless: +Cr +Ni +C
- Invar: + Ni

Aluminium base — light alloys
- Duralumin: Cu, Mg, Mn
- Magnalium: + Mg

Other bases — no copper, no iron
- Solder: Pb + Sn
- Nichrome: Ni+Cr+Fe
- Amalgam: any + Hg

ASKED IN RAS

Documented repeats: 'stainless steel is an alloy of iron with chromium and nickel', 'german silver is an alloy of copper, zinc and nickel', and the full alloy matching list with brass, bronze and duralumin. Note also that copper is the base metal of brass, bronze, german silver, gunmetal and bell metal - five of the alloys in the list.

Carbon, its allotropes and the fuels you use at home

Carbon is unique for two reasons. It has four bonds to give (tetravalency) and it can link to itself endlessly (catenation) - which is why there are millions of carbon compounds and only a few hundred thousand of everything else put together. The same pure carbon, arranged differently, gives the hardest substance known and one of the softest. Those different forms are the allotropes, and they are asked every year.

Allotropes of carbon - same element, opposite properties

Allotrope	Structure	Property and use
Diamond	Each carbon bonded to FOUR others in a rigid three-dimensional network	Hardest natural substance; does NOT conduct electricity; cutting, drilling, jewellery
Graphite	Flat layers that slide over each other, with one free electron per atom	Soft and slippery - a lubricant; CONDUCTS electricity - electrodes; pencil lead, mixed with clay
Fullerene (C60)	Sixty carbon atoms in a hollow football shape - buckminsterfullerene	Nobel Prize in Chemistry 1996 to Curl, Kroto and Smalley
Graphene	A single one-atom-thick layer of graphite	Thinnest and among the strongest materials known; Nobel Prize in Physics 2010 to Geim and Novoselov
Carbon nanotubes	Graphene sheets rolled into cylinders	Extremely strong and conducting; electronics and composites

- Amorphous forms of carbon: coal, coke, charcoal, lamp black. Activated charcoal ADSORBS gases and colouring matter - used in gas masks and water purifiers.
- Hydrocarbons: alkanes have only single bonds (methane, ethane), alkenes have a double bond (ethene), alkynes have a triple bond (ethyne, that is acetylene).
- Calcium carbide with water gives ACETYLENE - used in oxy-acetylene welding and misused for the artificial ripening of fruit, which is banned in India.
- ETHYLENE (ethene) is the natural ripening hormone of fruit - the safe and legal one. Do not confuse it with acetylene.
- Methanol is poisonous and causes blindness and death, which is why hooch tragedies happen. Ethanol is the alcohol in drinks; denatured alcohol is ethanol deliberately made undrinkable.
- Esters give fruits and perfumes their pleasant smell; boiling an ester or a fat with alkali gives soap, and that reaction is saponification.
- Calorific value is the heat released per unit mass of fuel burnt. HYDROGEN has the highest calorific value of any fuel, about 150 kJ per gram.
- A good fuel has a high calorific value, a moderate ignition temperature, little ash and smoke, and is easy to store and carry.
- LPG is stored as a liquid under pressure in the cylinder and leaves as a gas - that is what the L stands for.
- Petroleum refining separates the fractions in order: petroleum gas, petrol, kerosene, diesel, lubricating oil, paraffin wax, bitumen.
- Carbon monoxide is the killer produced by incomplete burning in a closed room - it binds to haemoglobin far more strongly than oxygen does.

CURRENT LINK

Current link, as of September 2026: the Ethanol Blended Petrol Programme is the fuel story to carry into the exam. India brought its 20 per cent blending target forward from 2030 to 2025-26 and reached the E20 level in 2025, and E20 petrol is now the standard grade at pumps. The ethanol comes from sugarcane molasses and from surplus grain such as maize and damaged foodgrain, and the stated aims are a smaller crude oil import bill, lower vehicular emissions and better prices for cane growers. Pair it with the National Green Hydrogen Mission when a question asks about alternative fuels.

Fuel gases and what they are made of

Fuel gas	Chief constituent	Note
CNG (compressed natural gas)	Mainly METHANE	The cleanest of the common vehicle fuels
LPG (liquefied petroleum gas)	Mainly BUTANE and propane	Ethyl mercaptan is added as an odorant so that a leak can be smelt
Biogas or gobar gas	Mainly methane	From anaerobic digestion of dung and waste
Water gas	Carbon monoxide + hydrogen	-
Producer gas	Carbon monoxide + nitrogen	-
Coal gas	Hydrogen + methane + carbon monoxide	-

ASKED IN RAS

Asked in RAS Pre 2021 and repeated - the main constituent of CNG is methane, while LPG is mainly butane and propane. Asked in RAS Pre 2016 - calcium carbide plus water gives acetylene, which is exactly why its use for ripening fruit is banned.

Polymers, plastics, soaps and the everyday materials – cement, glass, fertilisers

A polymer is a long chain built by joining one small unit over and over. Nature makes them – cellulose, starch, protein, natural rubber. Chemistry copies the trick to make plastics and synthetic fibres. The one classification RPSC always wants is whether the plastic can be melted and reshaped, or whether it has set for good.

Thermoplastic vs thermosetting plastic

Thermoplastic	Thermosetting plastic
Softens on heating and can be remoulded again and again	Sets permanently once moulded; heating only chars it
Long separate chains held by weak forces	Chains cross-linked into one rigid network
Polythene, PVC, polystyrene, nylon, PET, teflon	Bakelite, melamine
Buckets, bottles, pipes, non-stick coatings	Electrical switches and plugs, crockery, handles of cooking pans
Easier to recycle	Cannot be remelted, so hard to recycle

- Bakelite, made by Leo Baekeland, was the first fully synthetic plastic. Teflon (PTFE) is the non-stick coating on pans.
- Nylon is a synthetic NON-cellulosic fibre - it comes from coal, water and air, not from any plant. Rayon is regenerated cellulose, which is why it is called artificial silk.
- Terylene or Dacron is a polyester; PET, the plastic of drinking-water bottles, belongs to the same family.
- Vulcanisation - heating natural rubber with SULPHUR, discovered by Charles Goodyear - makes rubber harder, stronger and more elastic. Tyres depend on it.
- India banned identified single-use plastic items from 1 July 2022. Biodegradable plastics such as PLA and PHBV are the substitutes being promoted.
- Cotton, jute, silk and wool are natural fibres; nylon, polyester and acrylic are synthetic.
- Cement is made from limestone and clay. Gypsum, about 2 to 3 per cent, is added to SLOW DOWN the setting - not to speed it up.
- Concrete = cement + sand + gravel + water. Reinforced cement concrete adds steel bars.
- Ordinary glass is mainly silica, SiO_2. Borosilicate or Pyrex glass resists heat and sudden temperature change. Cobalt oxide colours glass blue; adding lead gives crystal glass.
- Fertilisers: urea, $CO(NH_2)_2$, has the highest nitrogen content among solid fertilisers, about 46 per cent. DAP gives nitrogen and phosphorus, muriate of potash is potassium chloride, superphosphate gives phosphorus.
- Explosives: gunpowder = potassium nitrate + charcoal + sulphur; TNT = trinitrotoluene; RDX = cyclonite; dynamite = nitroglycerine soaked in kieselguhr, invented by Alfred Nobel.

How soap actually removes grease – the micelle story

A soap molecule has a long oily TAIL and a water-loving HEAD → Tails bury themselves in the grease, heads face the water → The grease is caged inside a ball called a MICELLE → Soap also lowers the surface tension of water so it wets the cloth properly → Rinsing carries the micelles away with the dirt locked inside

- Soaps are sodium or potassium salts of long-chain fatty acids, made by saponification – boiling oil or fat with alkali. Potassium soaps are the soft ones.
- Soap does NOT lather in hard water: the calcium and magnesium salts in hard water form an insoluble curdy scum and the soap is simply wasted.
- Synthetic detergents are alkyl benzene sulphonates. They work perfectly well in hard water, which is the whole reason they were invented.
- The drawback of detergents is that branched-chain versions resist bacterial breakdown and cause foam in rivers; soaps are biodegradable.
- Temporary hardness is due to bicarbonates and is removed by boiling; permanent hardness is due to sulphates and chlorides and needs washing soda or an ion exchanger.

Batteries, fuel cells, catalysts and the gases around you

A cell turns chemical energy straight into electrical energy. A primary cell such as the dry cell is used once and thrown away; a secondary cell can be recharged by pushing current back through it. A fuel cell is different from both - it is not a store at all, it keeps producing electricity for as long as you keep feeding it fuel.

- Dry cell - a zinc-carbon primary cell; the zinc case is the negative electrode and the carbon rod the positive.
- Lead-acid storage battery - lead, lead dioxide and sulphuric acid; this is the rechargeable battery in cars and inverters.
- Lithium-ion cells power mobile phones, laptops and electric vehicles. The 2019 Nobel Prize in Chemistry went to Goodenough, Whittingham and Yoshino for developing them.
- A fuel cell combines hydrogen and oxygen to give electricity directly, with WATER as the only product - no combustion, no smoke. Fuel cells powered the Apollo spacecraft.
- Air is about 78 per cent nitrogen and 21 per cent oxygen by volume, with argon and carbon dioxide making up most of the rest.
- Argon, mixed with nitrogen, fills an ordinary incandescent bulb - an inert filling stops the hot filament from burning away.
- Helium fills balloons and airships because it is light and, unlike hydrogen, non-inflammable. It is also in the breathing mixture of deep-sea divers.
- Neon glows red-orange in advertising signs. Ozone, O3, in the stratosphere absorbs harmful ultraviolet radiation.
- Carbon dioxide is used in fire extinguishers because it is heavier than air and does not support burning; as dry ice it is a refrigerant.
- Hydrogen is the lightest element known; nitrogen provides an inert atmosphere for storing food and for filling tyres.

Industrial process to catalyst - a standing matching question

Process	Product	Catalyst
Haber process	Ammonia	Finely divided IRON
Contact process	Sulphuric acid	Vanadium pentoxide, V_2O_5
Hydrogenation of vegetable oil into vanaspati	Solid fat	NICKEL
Ostwald process	Nitric acid	Platinum

CURRENT LINK

Current link, as of September 2026: the 2025 Nobel Prize in Chemistry went to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for metal-organic frameworks (MOFs) - crystalline materials with enormous internal pore space, being developed to harvest water from desert air, capture carbon dioxide and store hydrogen. Remember the phrase 'metal-organic frameworks' with those three names. Alongside it, India's National Green Hydrogen Mission, approved in January 2023 with an outlay of about Rs 19,744 crore, targets 5 million tonnes of green hydrogen a year by 2030; green hydrogen is made by electrolysis of water using renewable electricity.

Drugs, food adulteration, radioisotopes and the master table of common names

Drugs are asked by class, not by chemistry. You need to know what each class DOES and one or two examples of it. The single most exploited pair is antiseptic against disinfectant - both kill germs, but one goes on you and the other goes on the floor.

- ANALGESIC - relieves pain. Aspirin and paracetamol are ordinary analgesics; morphine, obtained from opium, is a NARCOTIC analgesic.
- ANTIPYRETIC - brings down fever. Paracetamol is both analgesic and antipyretic.
- ANTIBIOTIC - kills or checks micro-organisms inside the body. Penicillin was discovered by Alexander Fleming in 1928.
- ANTACID - neutralises excess acid in the stomach: milk of magnesia and baking soda.
- ANTIHISTAMINE for allergy; TRANQUILLISER for anxiety and sleeplessness; antifertility drugs for conception control.
- Antibiotics do nothing against viruses. Overuse of antibiotics is exactly what breeds antimicrobial resistance.
- Food adulteration to memorise: argemone seed oil in mustard oil, which causes epidemic dropsy; metanil yellow in turmeric and pulses; brick powder in chilli powder; chicory in coffee; melamine in milk; papaya seeds in black pepper; chalk powder in wheat flour.
- Common preservatives: sodium benzoate and sodium metabisulphite, plus the traditional salt, sugar, oil and vinegar.
- FSSAI, the Food Safety and Standards Authority of India, set up under the Food Safety and Standards Act 2006, is the regulator. AGMARK certifies agricultural produce, BIS/ISI industrial goods, Hallmark gold.
- Iodised salt prevents goitre; fortification adds iron, vitamin A and vitamin D to staple foods.
- Aqua regia is 3 parts concentrated hydrochloric acid to 1 part concentrated nitric acid, and it dissolves even gold and platinum - the name means 'royal water' because it attacks the royal metals.

Classes of drugs – what each one does

Classes of drugs

Analgesic
relieves pain
- Aspirin, paracetamol
- Morphine = narcotic
- From opium

Antipyretic
brings fever down
- Paracetamol does both
- Fever, not pain

Antibiotic
kills micro-organisms
- Penicillin – Fleming
- Discovered in 1928
- No use on viruses

Antacid
neutralises stomach acid
- Milk of magnesia
- Baking soda
- A mild base only

Antiseptic vs disinfectant – the exact distinction that is asked

Antiseptic	Disinfectant
Applied to LIVING tissue – skin, wounds, mouth	Applied to NON-LIVING objects – floors, drains, instruments
Dettol, Savlon, tincture of iodine, boric acid, dilute hydrogen peroxide	Phenol in higher concentration, chlorine, formalin, bleaching powder
Mild, because it must not harm the tissue	Strong, and would harm living tissue
Dilute phenol works as an antiseptic	Concentrated phenol is a disinfectant – same chemical, different strength

Radioisotope to use

Radioisotope	Used for
Carbon-14	Dating of fossils and archaeological material
Cobalt-60	Radiotherapy for cancer
Iodine-131	Diagnosis and treatment of thyroid disorders
Phosphorus-32	Agricultural research and leukaemia studies
Sodium-24	Tracing blood circulation and finding leaks in underground pipelines
Uranium-235	Fuel for nuclear fission

MASTER TABLE – common name to formula, part 1

Common name	Chemical name	Formula
Baking soda	Sodium hydrogencarbonate	$NaHCO_3$
Washing soda	Sodium carbonate decahydrate	$Na_2CO_3.10H_2O$
Caustic soda	Sodium hydroxide	$NaOH$
Caustic potash	Potassium hydroxide	KOH
Quicklime	Calcium oxide	CaO
Slaked lime	Calcium hydroxide	$Ca(OH)_2$
Limestone, marble, chalk	Calcium carbonate	$CaCO_3$
Gypsum	Calcium sulphate dihydrate	$CaSO_4.2H_2O$
Plaster of Paris	Calcium sulphate hemihydrate	$CaSO_4.(1/2)H_2O$
Bleaching powder	Calcium oxychloride	$CaOCl_2$
Milk of magnesia	Magnesium hydroxide	$Mg(OH)_2$
Heavy water	Deuterium oxide	D_2O

MASTER TABLE - common name to formula, part 2

Common name	Chemical name	Formula
Blue vitriol	Copper sulphate pentahydrate	$CuSO_4.5H_2O$
Green vitriol	Ferrous sulphate heptahydrate	$FeSO_4.7H_2O$
White vitriol	Zinc sulphate heptahydrate	$ZnSO_4.7H_2O$
Oil of vitriol	Concentrated sulphuric acid	H_2SO_4
Epsom salt	Magnesium sulphate heptahydrate	$MgSO_4.7H_2O$
Glauber's salt	Sodium sulphate decahydrate	$Na_2SO_4.10H_2O$
Hypo, the photographic fixer	Sodium thiosulphate	$Na_2S_2O_3.5H_2O$
Dry ice	Solid carbon dioxide	CO_2
Laughing gas	Nitrous oxide	N_2O
Marsh gas	Methane	CH_4
Sal ammoniac	Ammonium chloride	NH_4Cl
Saltpetre	Potassium nitrate	KNO_3
Borax	Sodium tetraborate decahydrate	$Na_2B_4O_7.10H_2O$

Common name	Chemical name	Formula
Potash alum	Potassium aluminium sulphate	$K_2SO_4.Al_2(SO_4)_3.24H_2O$

ASKED IN RAS

Asked in RAS Pre 2016 and repeated - the formula of baking soda is $NaHCO_3$, and vinegar is a dilute solution of acetic acid. Asked in RAS Pre 2023 - morphine is a narcotic analgesic. Asked in RAS Pre 2018 - nylon is a synthetic non-cellulosic fibre. The common-name-to-formula matching set is a documented repeat, so both tables above are worth a full revision pass.

YAAD RAKHO

Remember the vitriols by colour: BLUE vitriol is copper, GREEN vitriol is iron (ferrous), WHITE vitriol is zinc, and OIL of vitriol is sulphuric acid itself. Colour to metal - that is the whole question.

REVISION RECAP

1. Sublimation - camphor, naphthalene, iodine, ammonium chloride, dry ice. Plasma is the fourth state, the Bose-Einstein condensate the fifth.
2. Mercury is the only liquid metal, bromine the only liquid non-metal, graphite the non-metal that conducts.
3. Thomson - electron; Chadwick - neutron; Rutherford - the nucleus. Isotopes share atomic number, isobars share mass number.
4. Mendeleev used atomic MASS; the modern periodic law of Moseley uses ATOMIC NUMBER. Fluorine is the most electronegative element.
5. Formic - ant; acetic - vinegar; citric - lemon; lactic - curd; oxalic - tomato; tartaric - tamarind; malic - apple; HCl - gastric juice.
6. pH 7 neutral, blood 7.35-7.45, tooth decay below 5.5, acid rain below about 5.6. Bee sting acidic so use baking soda; wasp sting alkaline so use vinegar.
7. Rust = $Fe_2O_3.xH_2O$ and needs oxygen AND moisture. Galvanisation is zinc, tinning or kalai is tin, stainless steel resists rust because of chromium.
8. Brass = Cu + Zn. Bronze = Cu + Sn. German silver = Cu + Zn + Ni with no silver in it. Gunmetal = Cu + Sn + Zn. Bell metal = Cu + Sn about 78:22.

9. Stainless steel = Fe + Cr about 18% + Ni about 8% + C. Duralumin = Al + Cu + Mg + Mn. Solder = Pb + Sn. Amalgam always contains mercury.
10. Bauxite - aluminium; haematite and magnetite - iron; galena - lead; cinnabar - mercury; zinc blende - zinc; copper pyrites - copper; pitchblende - uranium; monazite - thorium.
11. Diamond has four bonds per atom and does not conduct; graphite has sliding layers, one free electron, so it conducts and lubricates. Then fullerene C60, graphene, nanotubes.
12. CNG is methane; LPG is butane and propane with ethyl mercaptan for smell; water gas is CO + H2; producer gas is CO + N2. Hydrogen has the highest calorific value.
13. Calcium carbide with water gives acetylene, banned for ripening; ethylene is the natural ripening hormone.
14. Thermoplastics remould - polythene, PVC, nylon, PET, teflon. Thermosetting sets for good - bakelite, melamine. Vulcanisation is rubber plus sulphur.
15. Soap fails in hard water and forms scum; detergents work in hard water. Cleaning happens by micelle formation and by lowering surface tension.
16. Gypsum is added to cement to SLOW the setting. Glass is mainly silica; Pyrex is borosilicate. Urea carries about 46 per cent nitrogen.
17. Catalysts: Haber - iron; contact process - V2O5; vanaspati - nickel; Ostwald - platinum.
18. Antiseptic goes on living tissue (Dettol, tincture of iodine); disinfectant on non-living things (phenol, chlorine). Penicillin - Fleming, 1928. Morphine is a narcotic analgesic.
19. Argemone oil in mustard oil causes epidemic dropsy; metanil yellow adulterates turmeric; melamine adulterates milk. FSSAI is the regulator.
20. Carbon-14 dating, cobalt-60 cancer therapy, iodine-131 thyroid, sodium-24 blood flow and pipeline leaks. Aqua regia = 3 parts HCl to 1 part HNO3.